

University of Cape Town

Bayesian Computational Methods

Assignment

Bayesian Analysis Assignment

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Introduction

Question a

Each of the n birds is caught exactly once. The capture day of bird i is denoted by t_i and is treated as fixed. Every bird moults at the same constant rate, with moult lasting τ days for all birds. The start of moult for bird i, T_i, is an unobserved random variable, where

$$T_i \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \sigma^2), \qquad f_T(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right), \qquad i = 1, \dots, n. \tag{1}$$

Let $Z = \frac{T_i - \mu}{\sigma} \sim \mathrm{N}(0,1)$ and let Φ denote the corresponding CDF. In the Type 1 model, only the status ($s_i \in \{1,2,3\}$) of each bird is observed.

For $s_i = 3$, bird i has not yet started moult:

$$P(t_i) = \Pr(T_i > t_i) = \Pr\left(Z > \frac{t_i - \mu}{\sigma}\right) = 1 - \Phi\left(\frac{t_i - \mu}{\sigma}\right). \tag{2}$$

For $s_i = 2$, bird i is currently moulting:

$$Q(t_i) = \Pr(t_i - \tau \le T_i \le t_i) = \Pr\left(\frac{t_i - \tau - \mu}{\sigma} \le Z \le \frac{t_i - \mu}{\sigma}\right) = \Phi\left(\frac{t_i - \mu}{\sigma}\right) - \Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right). \tag{3}$$

For $s_i = 1$, bird i has finished moulting:

$$R(t_i) = \Pr(T_i \le t_i - \tau) = \Pr\left(Z \le \frac{t_i - \tau - \mu}{\sigma}\right) = \Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right). \tag{4}$$

Given t_i, the status s_i is a single categorical draw with probabilities $P(t_i), Q(t_i)$ and $R(t_i)$, given in Equation 2, Equation 3 and Equation 4. Thus, the conditional probability of s_i can be written as

$$\Pr(s_i \mid t_i, \mu, \sigma^2, \tau) = P(t_i)^{\mathrm{I}\{s_i=3\}}\, Q(t_i)^{\mathrm{I}\{s_i=2\}}\, R(t_i)^{\mathrm{I}\{s_i=1\}}. \tag{5}$$

Since the birds are independent, the likelihood is the product of Equation 5 over all n birds

$$\begin{aligned} L(\boldsymbol{s}, \boldsymbol{t}, \mu, \sigma^2, \tau) &= \prod_{i=1}^{n} P(t_i)^{\mathrm{I}\{s_i=3\}}\, Q(t_i)^{\mathrm{I}\{s_i=2\}}\, R(t_i)^{\mathrm{I}\{s_i=1\}} \\ &= \prod_{i=1}^{n} \left[1 - \Phi(\frac{t_i - \mu}{\sigma})\right]^{\mathrm{I}\{s_i=3\}} \left[\Phi(\frac{t_i - \mu}{\sigma}) - \Phi(\frac{t_i - \tau - \mu}{\sigma})\right]^{\mathrm{I}\{s_i=2\}} \left[\Phi(\frac{t_i - \tau - \mu}{\sigma})\right]^{\mathrm{I}\{s_i=1\}}. \end{aligned} \tag{6}$$

On any capture day t, the population consists of birds which belong to three different states.

Birds have either not yet started to moult, are undergoing moult or have completed moult. These states occur in proportions $P(t), Q(t)$ and $R(t)$, which depend on t through μ, σ^2 and τ. Thus, the moult index has a mixture distribution with a point mass at 0 and 1 with weights $P(t)$ and $R(t)$, respectively, as well as a continuous component on $(0, 1)$ with weight $Q(t)$. In the Type 1 model, the moult index is not observed. We only observe the state that bird i belongs to. Thus, each bird contributes the weight of the state it belongs to $P(t_i)$, $Q(t_i)$ or $R(t_i)$, to Equation 6.

Question b

In the Type 2 model, the moult scores (y_i) for birds that are currently moulting are observed. A Gibbs sampler is developed, using data augmentation, by treating the unobserved start dates of the birds that have not yet started moulting and of those that have finished moulting as latent variables. The derivations of the Type 2 likelihood and the conditional distributions are given in Appendix B.

Definitions

The birds are grouped according to their states. Let $\mathcal{F} = \{i : s_i = 1\}$ denote the set of birds that have already moulted, $\mathcal{M} = \{i : s_i = 2\}$ denote the set of birds that are undergoing moult and $\mathcal{N} = \{i : s_i = 3\}$ denote the set of birds that have not yet started to moult. Let $m = |\mathcal{M}|$ be the number of birds currently moulting.

For a bird undergoing moult, the linear moult assumption gives $T_i = t_i - y_i\tau$ and the probability density function of its observed moult score is

$$f_y(y_i) = \tau f_T(t_i - y_i\tau), \qquad 0 < y_i < 1. \tag{7}$$

The start dates T_i for $i \in \mathcal{F} \cup \mathcal{N}$ are treated as latent variables. The prior distributions are

$$\mu \sim \text{Uniform}(100, 160), \quad \sigma^2 \sim \text{IG}(1, 1), \quad \tau \sim \text{Uniform}(50, 180). \tag{8}$$

The following quantities are used in the conditional distributions below

$$\begin{aligned} \bar{T} &= \frac{1}{n}\sum_{i=1}^{n} T_i, \quad & S &= \sum_{i=1}^{n}(T_i - \mu)^2, & h &= \min_{i\in\mathcal{F}}(t_i - T_i), \\ \beta &= \frac{\sum_{i\in\mathcal{M}} y_i^2}{2\sigma^2}, & \gamma &= \frac{\sum_{i\in\mathcal{M}} y_i(t_i - \mu)}{\sigma^2}, & \bar{t}_y &= \frac{\sum_{i\in\mathcal{M}} y_i t_i}{\sum_{i\in\mathcal{M}} y_i}, \end{aligned} \tag{9}$$

where $T_i = t_i - y_i\tau$ is used for $i \in \mathcal{M}$.

Conditional distributions

The latent start dates follow the same truncated normal distributions as in the Type 1 Gibbs sampler

$$\begin{aligned} T_i \mid \cdot &\sim \mathrm{N}(\mu, \sigma^2)\, \mathrm{I}\{T_i \leq t_i - \tau\}, \quad i \in \mathcal{F}, \\ T_i \mid \cdot &\sim \mathrm{N}(\mu, \sigma^2)\, \mathrm{I}\{T_i > t_i\}, \qquad i \in \mathcal{N}. \end{aligned} \tag{10}$$

The full conditional distributions of μ and σ^2 are

$$\mu \mid \cdot \sim \mathrm{N}\left(\bar{T}, \frac{\sigma^2}{n}\right) \mathrm{I}\{100 \leq \mu \leq 160\}, \tag{11}$$

$$\sigma^2 \mid \cdot \sim \mathrm{IG}\left(\frac{n}{2} + 1,\ 1 + \frac{S}{2}\right). \tag{12}$$

Equation 12 differs from the Type 1 conditional given in the assignment, which corresponds to the prior $\pi(\sigma^2) \propto 1/\sigma^2$ as opposed to $\mathrm{IG}(1, 1)$.

The full conditional distribution of τ is a truncated modified half-normal (MHN) distribution (Sun et al., 2021)

$$\tau \mid \cdot \sim \mathrm{MHN}(m + 1,\ \beta,\ \gamma)\, \mathrm{I}\{50 \leq \tau \leq \min(180, h)\}. \tag{13}$$

In the Type 1 model, τ appears only in the indicators and has a uniform conditional distribution. In the Type 2 model, the observed moult scores result in a non-uniformly shaped conditional distribution of τ. Sampling from Equation 13 is simplified by a further data augmentation step based on Lemma 5 of Sun et al. (2021), the form of which depends on the sign of γ. Since $y_i > 0$, $\gamma \geq 0$ if and only if $\mu \leq \bar{t}_y$. For the `sanderlings` data, $\bar{t}_y = 164.4$, which exceeds the upper prior bound of μ, so $\gamma > 0$ at every iteration. A latent variable W is therefore introduced such that

$$\begin{aligned} W \mid \tau &\sim \mathrm{Poisson}(\gamma\tau), \\ \tau^2 \mid W &\sim \mathrm{Gamma}\left(\frac{m + 1 + W}{2},\ \beta\right) I_{\{\tau \in V\}}. \end{aligned} \tag{14}$$

The gamma distribution is parameterised by shape and rate. The truncated gamma distribution is sampled exactly using the inverse CDF method and τ is obtained as the positive square root of the draw. The latent variable W is redrawn given the current τ at every iteration and then discarded, so the conditional distributions of the other parameters are unchanged. The corresponding augmentation for $\gamma < 0$, which is not needed for these data, is given in Appendix B.

The Gibbs sampling algorithm

At each step, the most recently sampled values of all other variables are used.

1. Initialise. Draw starting values $\mu \sim \text{Uniform}(100, 160)$, $\sigma^2 \sim \text{IG}(1, 1)$ and $\tau \sim \text{Uniform}(50, 180)$. Then draw T_i for $i \in \mathcal{F} \cup \mathcal{N}$ from Equation 10.
2. Repeat for K iterations:
   a. Draw T_i for $i \in \mathcal{F} \cup \mathcal{N}$ from Equation 10.
   b. Set $T_i = t_i - y_i\tau$ for $i \in \mathcal{M}$.
   c. Draw μ from Equation 11.
   d. Draw σ^2 from Equation 12.
   e. Compute h, β and γ from Equation 9.
   f. Draw W and then τ^2 from Equation 14. Set τ to the positive square root.
3. Output. Discard the burn-in iterations and keep the draws of (μ, σ^2, τ). The latent variables T_i and W are not kept.

Question c

We sample from the posterior distribution of the Type 3 model, using the likelihood given in the assignment and the priors in Equation 8, with a random walk Metropolis-Hastings algorithm. However, the steps of the random walk are drawn from a multivariate normal distribution, while none of our three parameters has support over the entire real line. The mean start date μ is constrained by its uniform prior to lie between 100 and 160, the duration of moult τ is constrained by its prior to lie between 50 and 180, and the variance σ^2 is strictly positive. We define the following transformations

$$\phi_1 = \text{logit}\left(\frac{\mu - 100}{60}\right), \qquad \phi_2 = \log \sigma^2, \qquad \phi_3 = \text{logit}\left(\frac{\tau - 50}{130}\right). \tag{15}$$

These transformations each have support over the entire real line, so the random walk is performed on the transformed scale $\boldsymbol{\phi} = (\phi_1, \phi_2, \phi_3)$. A proposal $\boldsymbol{\phi}^* = \boldsymbol{\phi} + \boldsymbol{e}$, with $\boldsymbol{e} \sim \text{N}(\mathbf{0}, \Sigma)$, is accepted with probability $\min\{1, \exp[\log p(\boldsymbol{\phi}^*) - \log p(\boldsymbol{\phi})]\}$, since the proposal is symmetric. When transforming back to the original parameters $\boldsymbol{\theta} = (\mu, \sigma^2, \tau)$, the Jacobian of the transformation must be included so that the algorithm converges to the correct posterior. With $p_j = \text{logit}^{-1}(\phi_j)$, the log target becomes

$$\begin{aligned} \log p(\boldsymbol{\phi}) &= \log L(\mu, \sigma^2, \tau; \boldsymbol{y}) + \log \pi(\boldsymbol{\theta}) + \log |J(\boldsymbol{\phi})|, \\ \log |J(\boldsymbol{\phi})| &= \log\{60\, p_1(1 - p_1)\} + \phi_2 + \log\{130\, p_3(1 - p_3)\}. \end{aligned} \tag{16}$$

Since we are unsure how our parameters are correlated, the first pilot run of 5 000 iterations draws the steps of the random walk independently, with $\Sigma = 0.05\, I_3$. The covariance matrix of the second half of these draws, scaled by $2.38^2/d$ with $d = 3$ (Roberts et al., 1997), is used as Σ in a second pilot run of 10 000 iterations. The final proposal covariance matrix is $2.38^2/3$ times the covariance of the second pilot run, and is held fixed for all retained chains (a jitter of $10^{-8} I_3$ is added for numerical stability). The tuning lowered the acceptance rate from 0.50 in the first pilot run to 0.22 in the second.

We run three chains of 50 000 iterations each, with seed 2026. The chains are initialised randomly, with μ and τ drawn uniformly over their prior ranges and σ^2 drawn log-uniformly

on $(10, 2000)$ to give dispersed starting values. The starting values and acceptance rates are given in Table 1.

Table 1: Random starting values and acceptance rates of the three Metropolis-Hastings chains.

Chain	$\mu^{(0)}$	$\sigma^{2(0)}$	$\tau^{(0)}$	Acceptance rate
1	113.29	583.19	117.06	0.22
2	125.39	20.93	151.60	0.22
3	148.43	715.56	140.47	0.22

The acceptance rates of approximately 0.22 are close to the optimal rate of 0.234 (Roberts et al., 1997). This suggests that the steps of our random walk are neither too small, which would result in high acceptance rates and slow exploration, nor too large, which would result in low acceptance rates from jumping into areas of low density, and that the covariance matrix was adequately tuned in the pilot runs.

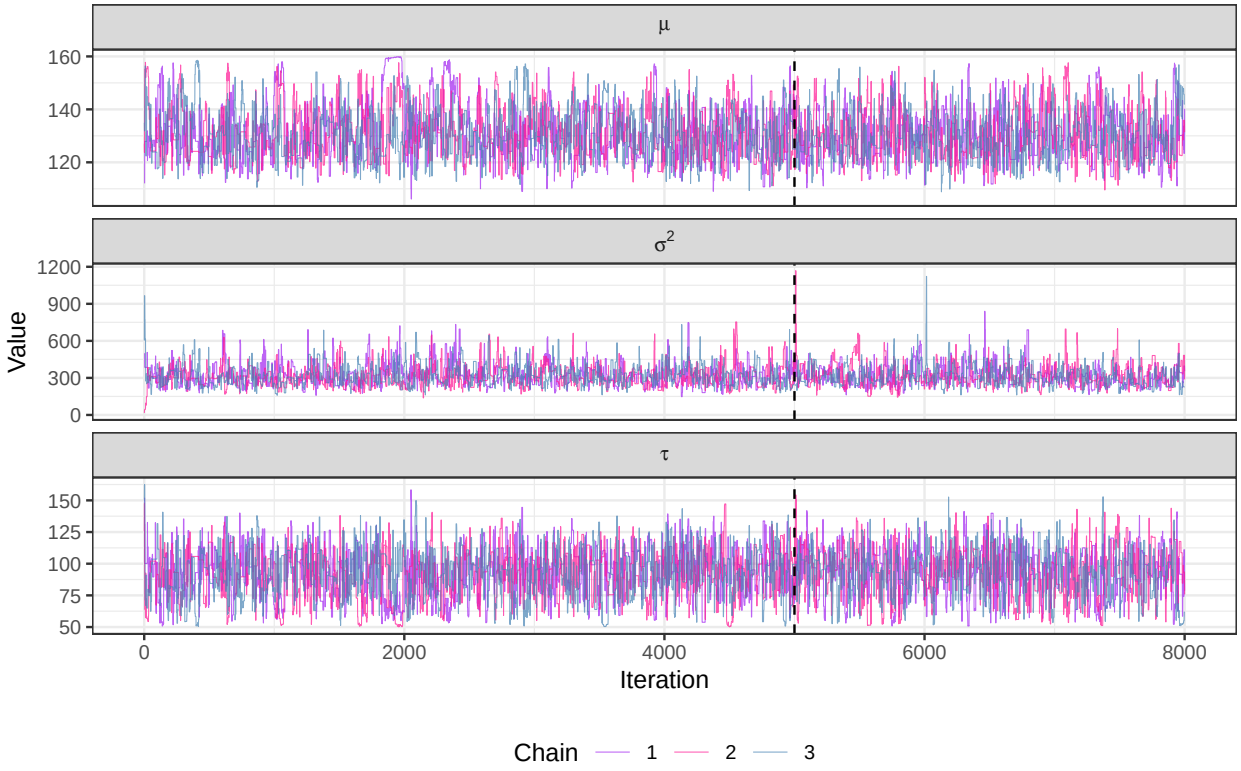


Figure 1: Trace plots of the three Metropolis-Hastings chains for the first 8 000 iterations. The burn-in is indicated by the dashed line.

The trace plots in Figure 1 show that the three chains, started from dispersed values, reach the same stationary region before the end of the burn-in, after which they overlap and mix well. A burn-in of 5 000 iterations is therefore discarded from each chain, leaving 45 000

draws per chain. The Gelman-Rubin statistic is $\hat{R} \leq 1.00$ for all parameters (Table 2), so we conclude that our sampler has converged. Successive draws are, however, autocorrelated (Figure 2), so the effective sample sizes, between 7 970 and 8 879, are much smaller than the 135 000 retained draws.

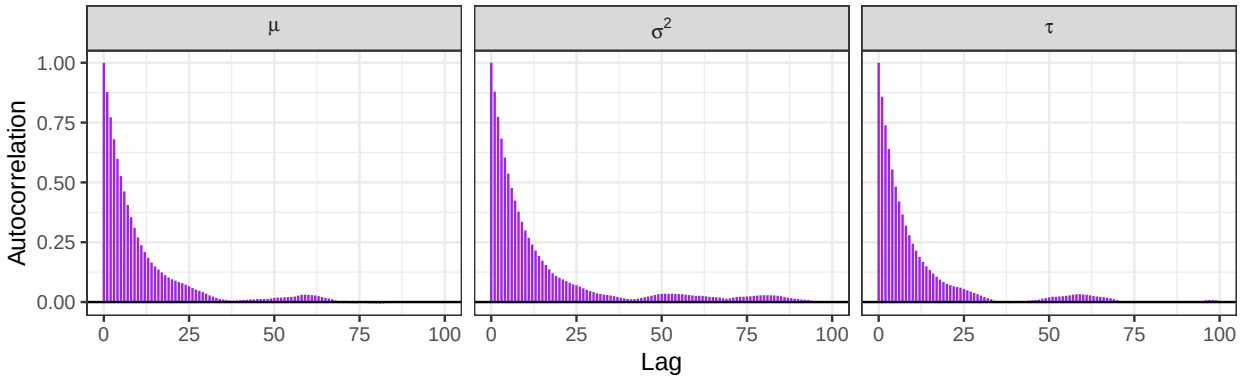


Figure 2: Autocorrelation of the retained draws of the first Metropolis-Hastings chain.

The posterior summaries and 95% credible intervals are given in Table 2, and the marginal and joint posterior distributions are shown in Figure 3 and Figure 4.

Table 2: Posterior summaries for the Type 3 model obtained using the tuned random walk Metropolis-Hastings algorithm (3 chains, 45 000 retained draws each).

	Mean	SD	2.5%	Median	97.5%	ESS	$\hat{R}$
μ	130.53	9.36	114.83	129.50	150.77	7970	1
σ^2	316.78	84.13	194.36	303.44	513.66	8024	1
τ	92.76	18.94	55.42	93.72	127.81	8879	1
σ	17.65	2.27	13.94	17.42	22.66	8046	1

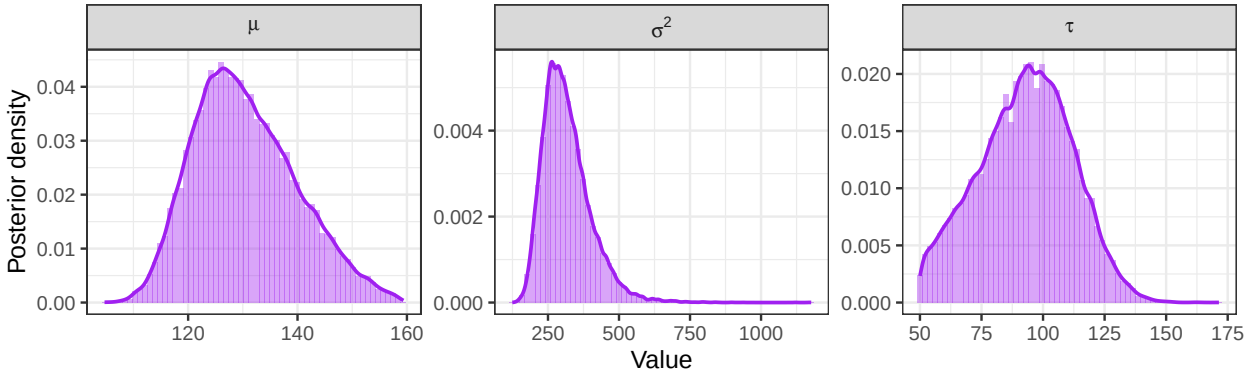


Figure 3: Marginal posterior distributions of the Type 3 model parameters from the Metropolis-Hastings draws.

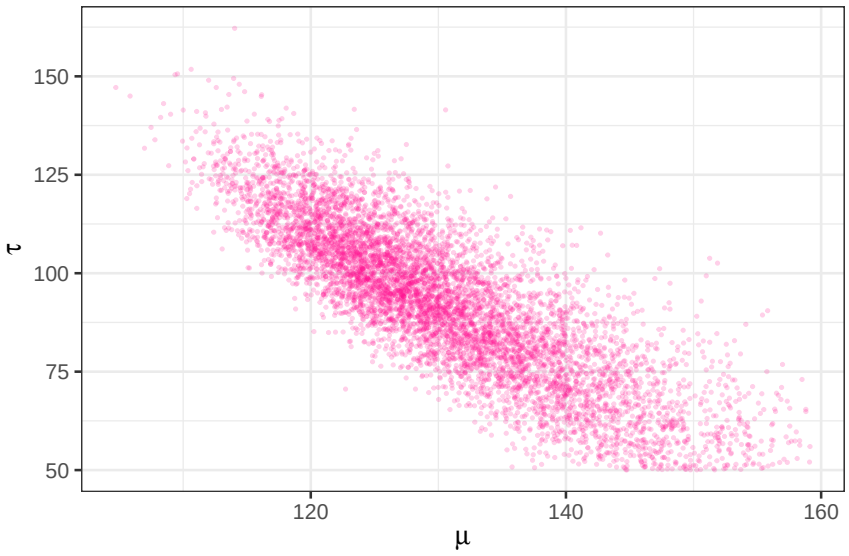


Figure 4: Joint posterior draws of the mean start date and duration of moult (every 20th draw shown).

The credible interval for μ is wide, covering a large part of the prior interval $(100, 160)$, and extends towards the upper prior bound, so the prior truncates the posterior of μ. Since 55 of the 66 moulting birds were caught on a single day (day 127), the moult scores carry little information to separate an early start with a long moult from a late start with a short moult. This explains the wide interval for μ and the strong negative correlation between μ and τ (ρ = -0.83, Figure 4), which is expected since $T_i = t_i - y_i\tau$: for the observed data, a longer duration implies earlier start dates and hence a smaller μ. The credible interval for τ is notably narrower than its prior. The parameters μ and σ^2 are broadly uncorrelated (ρ = 0.12), while τ is weakly correlated with σ^2 (ρ = 0.29), since the spread of the implied start dates $t_i - y_i\tau$ also depends on τ.

Question d

```
Running "C:/PROGRA~1/R/R-4.5.3/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.3.0'
gcc  -I"C:/PROGRA~1/R/R-4.5.3/include" -DNDEBUG   -I"C:/Users/isabe/AppData/
   Local/R/win-library/4.5/Rcpp/include/"  -I"C:/Users/isabe/AppData/Local/R/
   win-library/4.5/RcppEigen/include/"  -I"C:/Users/isabe/AppData/Local/R/win-
   library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/isabe/AppData/Local
   /R/win-library/4.5/BH/include" -I"C:/Users/isabe/AppData/Local/R/win-
   library/4.5/StanHeaders/include/src/"  -I"C:/Users/isabe/AppData/Local/R/
   win-library/4.5/StanHeaders/include/"  -I"C:/Users/isabe/AppData/Local/R/
   win-library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -I"C:/
   Users/isabe/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG
     -DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS
     -DUSE_STANC3 -DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION
   -D_HAS_AUTO_PTR_ETC=0  -include "C:/Users/isabe/AppData/Local/R/win-library
   /4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:
```

```
   /rtools45/x86_64-w64-mingw32.static.posix/include"      -O2 -Wall -std=
   gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but
    not for C
In file included from C:/Users/isabe/AppData/Local/R/win-library/4.5/RcppEigen
   /include/Eigen/Core:19,
                 from C:/Users/isabe/AppData/Local/R/win-library/4.5/RcppEigen
   /include/Eigen/Dense:1,
                 from C:/Users/isabe/AppData/Local/R/win-library/4.5/
   StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
                 from <command-line>:
C:/Users/isabe/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/
   Core/util/Macros.h:679:10: fatal error: cmath: No such file or directory
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-4.5.3/etc/x64/Makeconf:295: foo.o] Error 1
```

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 0.000129 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take
   1.29 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:     1 / 52000 [  0%]  (Warmup)
Chain 1: Iteration:  2001 / 52000 [  3%]  (Sampling)
Chain 1: Iteration:  7200 / 52000 [ 13%]  (Sampling)
Chain 1: Iteration: 12400 / 52000 [ 23%]  (Sampling)
Chain 1: Iteration: 17600 / 52000 [ 33%]  (Sampling)
Chain 1: Iteration: 22800 / 52000 [ 43%]  (Sampling)
Chain 1: Iteration: 28000 / 52000 [ 53%]  (Sampling)
Chain 1: Iteration: 33200 / 52000 [ 63%]  (Sampling)
Chain 1: Iteration: 38400 / 52000 [ 73%]  (Sampling)
Chain 1: Iteration: 43600 / 52000 [ 83%]  (Sampling)
Chain 1: Iteration: 48800 / 52000 [ 93%]  (Sampling)
Chain 1: Iteration: 52000 / 52000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 1.045 seconds (Warm-up)
Chain 1:                26.118 seconds (Sampling)
Chain 1:                27.163 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
Chain 2: Rejecting initial value:
Chain 2:   Log probability evaluates to log(0), i.e. negative infinity.
Chain 2:   Stan can't start sampling from this initial value.
Chain 2: Rejecting initial value:
Chain 2:   Log probability evaluates to log(0), i.e. negative infinity.
Chain 2:   Stan can't start sampling from this initial value.
Chain 2: Rejecting initial value:
Chain 2:   Log probability evaluates to log(0), i.e. negative infinity.
```

```
Chain 2:   Stan can't start sampling from this initial value.
Chain 2:
Chain 2: Gradient evaluation took 4.6e-05 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take
   0.46 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:     1 / 52000 [  0%]  (Warmup)
Chain 2: Iteration:  2001 / 52000 [  3%]  (Sampling)
Chain 2: Iteration:  7200 / 52000 [ 13%]  (Sampling)
Chain 2: Iteration: 12400 / 52000 [ 23%]  (Sampling)
Chain 2: Iteration: 17600 / 52000 [ 33%]  (Sampling)
Chain 2: Iteration: 22800 / 52000 [ 43%]  (Sampling)
Chain 2: Iteration: 28000 / 52000 [ 53%]  (Sampling)
Chain 2: Iteration: 33200 / 52000 [ 63%]  (Sampling)
Chain 2: Iteration: 38400 / 52000 [ 73%]  (Sampling)
Chain 2: Iteration: 43600 / 52000 [ 83%]  (Sampling)
Chain 2: Iteration: 48800 / 52000 [ 93%]  (Sampling)
Chain 2: Iteration: 52000 / 52000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 1.086 seconds (Warm-up)
Chain 2:                24.302 seconds (Sampling)
Chain 2:                25.388 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 4.8e-05 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take
   0.48 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:     1 / 52000 [  0%]  (Warmup)
Chain 3: Iteration:  2001 / 52000 [  3%]  (Sampling)
Chain 3: Iteration:  7200 / 52000 [ 13%]  (Sampling)
Chain 3: Iteration: 12400 / 52000 [ 23%]  (Sampling)
Chain 3: Iteration: 17600 / 52000 [ 33%]  (Sampling)
Chain 3: Iteration: 22800 / 52000 [ 43%]  (Sampling)
Chain 3: Iteration: 28000 / 52000 [ 53%]  (Sampling)
Chain 3: Iteration: 33200 / 52000 [ 63%]  (Sampling)
Chain 3: Iteration: 38400 / 52000 [ 73%]  (Sampling)
Chain 3: Iteration: 43600 / 52000 [ 83%]  (Sampling)
Chain 3: Iteration: 48800 / 52000 [ 93%]  (Sampling)
Chain 3: Iteration: 52000 / 52000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 1.083 seconds (Warm-up)
Chain 3:                33.351 seconds (Sampling)
Chain 3:                34.434 seconds (Total)
Chain 3:
```

Table 3: Posterior summaries for the Type 3 model obtained using Stan (NUTS), based on 3 chains of 50 000 post-warmup draws.

	Mean	SD	2.5%	Median	97.5%	ESS	$\hat{R}$
μ	130.75	9.53	115.05	129.71	151.50	41715	1
σ^2	317.93	85.52	194.22	303.70	524.47	63042	1
τ	92.50	19.15	55.25	93.35	128.13	45646	1
σ	17.68	2.30	13.94	17.43	22.90	65274	1

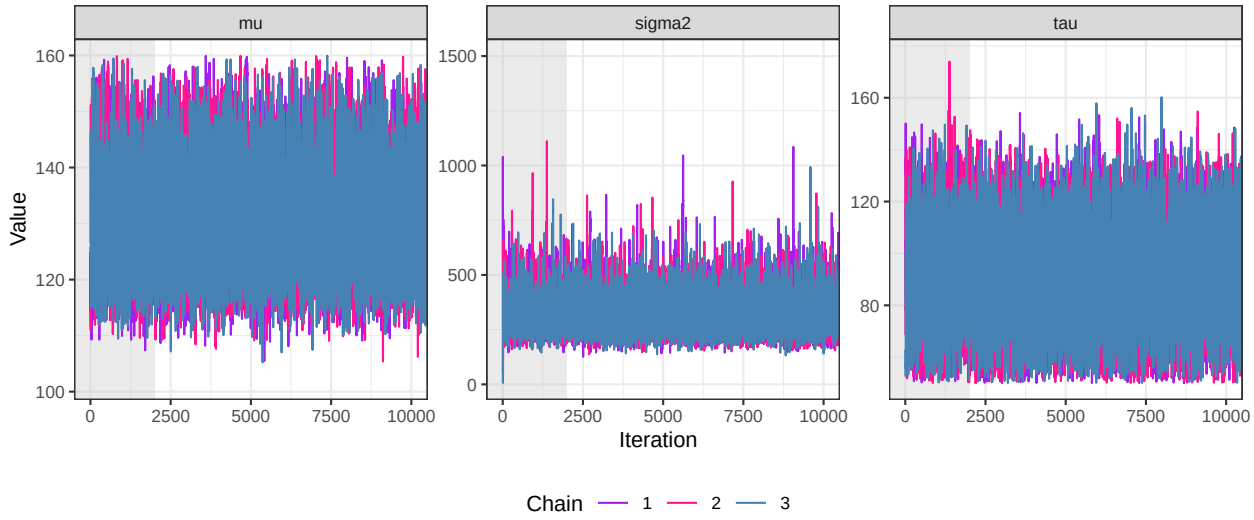


Figure 5: Trace plots of the three Stan chains for the first 4 000 iterations. The shaded region is the warmup period (2 000 iterations), which is discarded.

Appendix A: Question b derivations

The likelihood contributions of birds that have already moulted and birds that have not yet started to moult are the same as in the Type 1 model. The moulting birds contribute the density in Equation 7. Thus, the Type 2 likelihood is

$$L(\boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \mu, \sigma^2, \tau) = \prod_{i \in \mathcal{F}} R(t_i) \prod_{i \in \mathcal{M}} \tau f_T(t_i - y_i \tau) \prod_{i \in \mathcal{N}} P(t_i), \tag{17}$$

and the joint posterior distribution is

$$\pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}) \propto \prod_{i \in \mathcal{F}} R(t_i) \prod_{i \in \mathcal{M}} \tau f_T(t_i - y_i \tau) \prod_{i \in \mathcal{N}} P(t_i)\, \pi(\mu)\, \pi(\sigma^2)\, \pi(\tau), \tag{18}$$

with the prior distributions in Equation 8.

Data augmentation

The probabilities $R(t_i)$ and $P(t_i)$ in Equation 18 are integrals over the unobserved start date

$$\begin{aligned} R(t_i) &= \Pr(T_i \le t_i - \tau) = \int_{-\infty}^{\infty} f_T(T_i)\, \mathrm{I}\{T_i \le t_i - \tau\}\, dT_i, \\ P(t_i) &= \Pr(T_i > t_i) = \int_{-\infty}^{\infty} f_T(T_i)\, \mathrm{I}\{T_i > t_i\}\, dT_i. \end{aligned} \tag{19}$$

The status is fully determined by the start date (T_i), so each integrand in Equation 19 is the joint density of the start date and the bird's status

$$\begin{aligned} p(T_i, s_i = 1 \mid \mu, \sigma^2, \tau) &= f_T(T_i)\, \mathrm{I}\{T_i \le t_i - \tau\}, \\ p(T_i, s_i = 3 \mid \mu, \sigma^2) &= f_T(T_i)\, \mathrm{I}\{T_i > t_i\}. \end{aligned} \tag{20}$$

T_i for $i \in \mathcal{F} \cup \mathcal{N}$ is treated as a latent variable and samples are drawn from the augmented posterior

$$\begin{aligned} \pi(\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})}, \mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}) \propto & \prod_{i\in\mathcal{F}} f_T(T_i)\, \mathrm{I}\{T_i \le t_i - \tau\} \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \\ & \times \prod_{i\in\mathcal{N}} f_T(T_i)\, \mathrm{I}\{T_i > t_i\}\, \pi(\mu)\, \pi(\sigma^2)\, \pi(\tau). \end{aligned} \tag{21}$$

Integrating out each T_i in Equation 21 returns $R(t_i)$ or $P(t_i)$, so that

$$\int \pi(\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})}, \mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t})\, d\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})} = \pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}). \tag{22}$$

The posterior distribution of interest, Equation 18, is the marginal distribution of the augmented posterior. Thus, the draws of μ, σ^2 and τ from a sampler targeting the augmented posterior are draws from the required posterior. No latent variables are needed for the moulting birds, since $T_i = t_i - y_i\tau$ is a deterministic function of τ for $i \in \mathcal{M}$. Since $0 < y_i < 1$, the condition $t_i - \tau \le T_i \le t_i$ is always satisfied, so no indicator is required for these birds.

Each full conditional distribution is obtained from Equation 21 by keeping only the factors that involve the variable being updated. For each latent T_i, the only such factor is $f_T(T_i)$ multiplied by its indicator, which gives Equation 10.

Conditional distribution of T_i

The conditional distributions of the latent start dates are the same truncated normal distributions as in the Type 1 Gibbs sampler

$$\begin{aligned} T_i \mid \cdot &\sim \mathrm{N}(\mu, \sigma^2)\, \mathrm{I}\{T_i \le t_i - \tau\}, \quad i \in \mathcal{F}, \\ T_i \mid \cdot &\sim \mathrm{N}(\mu, \sigma^2)\, \mathrm{I}\{T_i > t_i\}, \qquad i \in \mathcal{N}. \end{aligned} \tag{23}$$

Conditional distribution of μ

From Equation 21, with $T_i = t_i - y_i\tau$ for the moulting birds,

$$\begin{aligned}
\pi(\mu \mid \cdot) &\propto \prod_{i\in\mathcal{F}} f_T(T_i) \prod_{i\in\mathcal{M}} f_T(t_i - y_i\tau) \prod_{i\in\mathcal{N}} f_T(T_i)\,\pi(\mu) \\
&= \prod_{i\in\mathcal{F}} f_T(T_i) \prod_{i\in\mathcal{M}} f_T(T_i) \prod_{i\in\mathcal{N}} f_T(T_i)\,\pi(\mu) \\
&= \prod_{i=1}^{n} f_T(T_i)\,\pi(\mu) \\
&= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(T_i-\mu)^2}{2\sigma^2}\right) \mathrm{I}\{100 \le \mu \le 160\} \\
&\propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(T_i-\mu)^2\right) \mathrm{I}\{100 \le \mu \le 160\}.
\end{aligned}$$

Expanding the sum of squares gives

$$\begin{aligned}
\sum_{i=1}^{n}(T_i-\mu)^2 &= \sum_{i=1}^{n}\left[(T_i-\bar{T}) + (\bar{T}-\mu)\right]^2 \\
&= \sum_{i=1}^{n}(T_i-\bar{T})^2 + 2(\bar{T}-\mu)\sum_{i=1}^{n}(T_i-\bar{T}) + n(\bar{T}-\mu)^2 \\
&= \sum_{i=1}^{n}(T_i-\bar{T})^2 + n(\bar{T}-\mu)^2,
\end{aligned}$$

since $\sum_{i=1}^{n}(T_i-\bar{T}) = 0$. The first term does not depend on μ. Thus,

$$\pi(\mu \mid \cdot) \propto \exp\left(-\frac{n}{2\sigma^2}(\mu-\bar{T})^2\right) \mathrm{I}\{100 \le \mu \le 160\}.$$

This is the kernel of the truncated normal distribution in Equation 11.

Conditional distribution of σ^2

From Equation 21, with $T_i = t_i - y_i\tau$ for the moulting birds and the IG(1, 1) prior $\pi(\sigma^2) \propto (\sigma^2)^{-2}e^{-1/\sigma^2}$,

$$\begin{aligned}
\pi(\sigma^2 \mid \cdot) &\propto \prod_{i\in\mathcal{F}} f_T(T_i) \prod_{i\in\mathcal{M}} f_T(t_i - y_i\tau) \prod_{i\in\mathcal{N}} f_T(T_i)\, \pi(\sigma^2) \\
&\propto \prod_{i=1}^{n} f_T(T_i)\, \pi(\sigma^2) \\
&\propto (\sigma^2)^{-n/2} \exp\left(-\frac{S}{2\sigma^2}\right) (\sigma^2)^{-2} \exp\left(-\frac{1}{\sigma^2}\right) \\
&= (\sigma^2)^{-(\frac{n}{2}+1)-1} \exp\left(-\frac{1+\frac{S}{2}}{\sigma^2}\right).
\end{aligned}$$

This is the kernel of the inverse-gamma distribution in Equation 12.

Conditional distribution of τ

From Equation 21, with $T_i = t_i - y_i\tau$ for the moulting birds

$$\begin{aligned}
\pi(\tau \mid \cdot) &\propto \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{F}} \mathrm{I}\{T_i \le t_i - \tau\}\, \pi(\tau) \\
&\propto \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{F}} \mathrm{I}\{T_i \le t_i - \tau\}\, \mathrm{I}\{50 \le \tau \le 180\}.
\end{aligned}$$

The moulting birds contribute

$$\prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \propto \tau^m \exp\left(-\frac{1}{2\sigma^2} \sum_{i\in\mathcal{M}} (t_i - \mu - y_i\tau)^2\right).$$

Expanding the square gives

$$\sum_{i\in\mathcal{M}} (t_i - \mu - y_i\tau)^2 = \sum_{i\in\mathcal{M}} (t_i - \mu)^2 - 2\tau \sum_{i\in\mathcal{M}} y_i(t_i - \mu) + \tau^2 \sum_{i\in\mathcal{M}} y_i^2,$$

where the first term does not depend on τ. Thus,

$$\prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \propto \tau^m \exp\left(-\beta\tau^2 + \gamma\tau\right).$$

Each indicator of a finished bird can be rearranged as $T_i \le t_i - \tau \iff \tau \le t_i - T_i$. The product of the indicators equals one only if this holds for every finished bird, so

$$\prod_{i\in\mathcal{F}} \mathrm{I}\{T_i \le t_i - \tau\} = \mathrm{I}\Big\{\tau \le \min_{i\in\mathcal{F}}(t_i - T_i)\Big\} = \mathrm{I}\{\tau \le h\}.$$

That is, the duration of moult cannot exceed the time between the start date and the capture date of any finished bird. Combining these terms with the prior gives

$$\pi(\tau \mid \cdot) \propto \tau^m \exp\left(-\beta\tau^2 + \gamma\tau\right) \mathrm{I}\{50 \leq \tau \leq \min(180, h)\}.$$

This is the kernel of the truncated MHN distribution in Equation 13, since the MHN density is proportional to $x^{\alpha-1}\exp(-\beta x^2 + \gamma x)$ for $x > 0$, with $\alpha = m + 1$.

Data augmentation for τ

For the `sanderlings` data, $\gamma > 0$ at every iteration, so only the Poisson augmentation is required. Multiplying the conditional distribution of τ by the Poisson probability of $W \mid \tau$ gives

$$\tau^m e^{-\beta\tau^2+\gamma\tau} \cdot \frac{e^{-\gamma\tau}(\gamma\tau)^W}{W!} \propto \tau^{m+W} e^{-\beta\tau^2},$$

so the term $e^{\gamma\tau}$ cancels. Changing variables from τ to τ^2, with $d\tau = d(\tau^2)/(2\tau)$,

$$\tau^{m+W} e^{-\beta\tau^2}\, d\tau \propto (\tau^2)^{\frac{m+1+W}{2}-1} e^{-\beta\tau^2}\, d(\tau^2),$$

which is the kernel of a Gamma $\left(\frac{m+1+W}{2}, \beta\right)$ distribution for τ^2. Since the indicator only involves τ, the truncation carries over to τ^2, which gives Equation 14.

W is drawn from its conditional given the current τ, followed by τ^2 given W. This two-step update leaves the conditional distribution of τ in Equation 13 invariant. If $\gamma < 0$, Lemma 5 of Sun et al. (2021) instead uses a generalised inverse Gaussian latent variable, which is not needed here.